Homework 10, Week 13 2023.11.29 Due 2022.12.04 补充 AJM† and others

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A particle of mass *m* performs a one-dimensional movement in the potential : $V(q) = c\, e^{\gamma q}$, with c and γ being both real constant.

Calculate $q(t)$ and $p(t)$ by use of the Hamilton-Jacobi method. Using a generating function of the type $U_2(q, P, t)$

Solution

$H = \dfrac{p^2}{2m} + c\, e^{\gamma q}$ 不显含时间，故： $S(q, P, t) = -E\, t + W(q, P)$

哈雅方程：$H\left(q, \dfrac{\partial S}{\partial q}\right) + \dfrac{\partial S}{\partial t} = 0 \implies \dfrac{1}{2m}\left(\dfrac{\partial W}{\partial q}\right)^2 + c\, e^{\gamma q} = E \implies \dfrac{\partial W}{\partial q} = \sqrt{2m}\,\sqrt{E - c\, e^{\gamma q}}$

$W = \sqrt{2m}\displaystyle\int \sqrt{E - c\, e^{\gamma q}}\, dq$ 注意 $E = \alpha = P$

$$Q = \frac{\partial S}{\partial \alpha} = \sqrt{\frac{m}{2}} \int \frac{dq}{\sqrt{\alpha - c\, e^{\gamma q}}} - t = \beta \qquad (\dagger)$$

$\implies Q = t + \beta$

令 $x = \sqrt{c}\; e^{\gamma q/2}$ $\quad dq = \dfrac{2}{\gamma}\dfrac{dx}{x}$ 代入 (†) 式 得

$$Q = \frac{\sqrt{2m}}{\gamma} \int \frac{1}{\sqrt{\alpha - x^2}} \frac{dx}{x} - t = \frac{\sqrt{m/(2\alpha)}}{\gamma} \int \frac{1}{\sqrt{1 - y/\alpha}} \frac{dy/\alpha}{y/\alpha} - t \qquad \text{with } y = x^2$$

$$= -\frac{\sqrt{2m/\alpha}}{\gamma} \operatorname{arctanh}\left(\sqrt{1 - x^2/\alpha}\right) - t = \beta$$

解得：$\sqrt{1 - x^2/\alpha} = -\tanh\left[\gamma\sqrt{\dfrac{\alpha}{2m}}\,(\beta + t)\right] \implies \dfrac{x^2}{\alpha} = 1 - \tanh^2\left[\gamma\sqrt{\dfrac{\alpha}{2m}}\,(\beta + t)\right]$

$$\implies q(t) = \frac{1}{\gamma} \ln\left[\frac{\alpha/c}{\cosh^2\left(\gamma\sqrt{\dfrac{\alpha}{2m}}\,(\beta + t)\right)}\right]$$

$p = \dfrac{\partial S}{\partial q} = \dfrac{\partial W}{\partial q} = \sqrt{2m}\,\sqrt{\alpha - c\, e^{\gamma q}}$ 上式的 $q(t)$ 代入即可

其中 α, β 由初始条件确定

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A particle of mass *m* performs a two-dimensional movement in the *x*-*y* plane in the potential: $V(x, y) = c\,(x - y)$, with c being a real constant.

Solve the Hamilton-Jacobi differential equation for the characteristic function $W(x, y, P_x, P_y)$. Use the initial conditions

$$x(t = 0) = y(t = 0) = 0, \quad \dot{x}(t = 0) = v_0 > 0, \quad \dot{y}(t = 0) = 0$$

Solution

$H = \dfrac{p_x^2 + p_y^2}{2m} + c\,(x - y)$ 不显含时间，故： $S(q, P, t) = -E\, t + W(q, P)$

哈雅方程：$H\left(q, \frac{\partial S}{\partial q}\right) + \frac{\partial S}{\partial t} = 0 \implies \frac{1}{2m}\left[\left(\frac{\partial W}{\partial x}\right)^2 + \left(\frac{\partial W}{\partial y}\right)^2\right] + c(x-y) = E$

分离变量：$\frac{1}{2m}\left(\frac{\partial W}{\partial x}\right)^2 + c\,x + \frac{1}{2m}\left(\frac{\partial W}{\partial y}\right)^2 - c\,y = E$ 令 $W = W_x(x, P_x, P_y) + W_y(x, P_x, P_y)$

$$\begin{cases} \frac{1}{2m}\left(\frac{\partial W_x}{\partial x}\right)^2 + c\,x = \alpha_1 \\ \frac{1}{2m}\left(\frac{\partial W_y}{\partial y}\right)^2 - c\,y = E - \alpha_1 \end{cases} \implies \begin{cases} \frac{dW}{dx} = \pm\sqrt{2m(\alpha_1 - c\,x)} \\ \frac{dW}{dy} = \pm\sqrt{2m(E - \alpha_1 + c\,y)} \end{cases} \implies \begin{cases} W_x = \mp\frac{1}{3mc}[2m(\alpha_1 - c\,x)]^{3/2} \\ W_y = \pm\frac{1}{3mc}[2m(E - \alpha_1 + c\,x)]^{3/2} \end{cases}$$

$$W = \mp\frac{\sqrt{2m}}{3mc}\left[(\alpha_x - c\,x)^{3/2} - (E - \alpha_x + c\,x)^{3/2}\right]$$

P_1, P_2 取为 α_1, E

$$\begin{cases} p_x = \frac{\partial W}{\partial x} = \pm\sqrt{2m(\alpha_1 - c\,x)} \\ p_y = \frac{\partial W}{\partial y} = \mp\sqrt{2m(E - \alpha_1 + c\,y)} \end{cases} \qquad \begin{cases} Q_1 = \frac{\partial S}{\partial \alpha_1} = \mp\frac{1}{c}\left[\sqrt{2m(\alpha_1 - c\,x)} + \sqrt{2m(E - \alpha_1 + c\,y)}\right] = \beta_x \\ Q_2 = \frac{\partial S}{\partial E} = -t \pm \frac{1}{c}\sqrt{2m(E - \alpha_1 + c\,y)}\Big] = \beta_2 \end{cases}$$

后两个方程解得：

$$y(t) = \frac{c}{2m}(t + \beta_2)^2 - \frac{E - \alpha_1}{c}, \quad x(t) = -\frac{c}{2m}(t + \beta_1 + \beta_2)^2 + \frac{\alpha_1}{c}$$

代入前两个方程得

$$p_x = \pm c\,(t + \beta_1 + \beta_2), \quad p_y = \mp c\,(t + \beta_2) \implies p_x = -c\,(t + \beta_1 + \beta_2), \quad p_y = c\,(t + \beta_2)$$

其中由 $x(t)$, $y(t)$ 确定 $p_x = m\,\dot{x}$, $p_y = m\,\dot{y}$ 的符号

接着由初条确定：$\beta_2 = 0$, $\beta_1 = -\frac{m\,v_0}{c}$, $\alpha_1 = \alpha_2 = \frac{1}{2}m\,v_0^2$

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Given the linear harmonic oscillator $H = \frac{p^2}{2m} + \frac{1}{2}m\,\omega_0^2\,q^2$. Perform in the theory of Hamilton-Jacobi theory a canonical transformation in such a manner the new coordinates (Q, P) are constant of motion. The generating function may be of the type $U(Q, p, t)$. Calculate $q(t)$ and $p(t)$ with the initial conditions: $p(t=0) = 0$, $q(t=0) = q_0 > 0$.

Solution

通常，人们选择生成函数 $U_2(q, P, t)$，见课件 例题 6.2.07，现在换一类生成函数为 $U(p, Q, t)$

$$q = -\frac{\partial U}{\partial p}, \quad P = -\frac{\partial U}{\partial Q}, \implies \text{哈雅方程：} H\left(-\frac{\partial S}{\partial p}, p, t\right) + \frac{\partial U}{\partial t} = 0, \quad U = -E\,t + W(p, Q)$$

$$\implies \frac{p^2}{2m} + \frac{1}{2}m\,\omega_0^2\left(\frac{\partial W}{\partial p}\right)^2 = E \implies \frac{dW}{dp} = \pm\frac{1}{m\,\omega_0}\sqrt{2mE - p^2} \implies W = \pm\frac{1}{m\,\omega_0}\int\sqrt{2mE - p^2}\,dp$$

$U(p, Q, t) = -E\,t + W(p, Q) = -E\,t \pm \frac{1}{m\,\omega_0}\int\sqrt{2mE - p^2}\,dp$ 含一个积分常数 $\alpha = E$

这时应令 $\alpha = E = Q$，—— 这是因为哈密顿主函数是 $U(p, Q, t)$ 的函数，Q 是运动常数，所以就令 $Q = \alpha = E$

$$P = -\frac{\partial U}{\partial Q} = -\frac{\partial U}{\partial E} = t \mp \frac{1}{\omega_0}\int\frac{dp}{\sqrt{2mE - p^2}} = t \mp \frac{1}{\omega_0}\arcsin\left(\frac{p}{\sqrt{2mE}}\right) = \beta = \text{const.}$$

$$p = \pm\sqrt{2mE}\,\sin(\omega_0\,t - \omega_0\,\beta)$$

$$q = -\frac{\partial U}{\partial p} = \mp \frac{1}{m\,\omega_0}\sqrt{2\,m\,E - p^2} \quad \text{代入上式的 } p \quad \Longrightarrow \quad q = \mp \frac{1}{\omega_0}\sqrt{\frac{2\,E}{m}}\,\cos(\omega_0\,t - \omega_0\,\beta)$$

初条：$t = 0$ 时，$\begin{cases} p(0) = 0 \\ q(0) = q_0 > 0 \end{cases} \quad \Longrightarrow \quad \begin{cases} \beta = 0 \\ q_0 = \sqrt{\dfrac{2\,E}{m\,\omega_0^2}} \end{cases}$

故：$\begin{cases} q(t) = q_0 \cos\omega_0\,t \\ p(t) = -m\,\omega_0\,q_0 \sin\omega_0\,t \end{cases}$

将 $p(t) = -m\,\omega_0\,q_0 \sin\omega_0\,t$ 代入哈密顿主函数，得

$$U(p,\,Q,\,t) = -E\,t - \frac{1}{m\,\omega_0}\int\sqrt{2\,m\,E - p^2}\;dp = -E\,t + 2\,E\int\cos^2\omega_0\,t\;dt = E\int\left(2\cos^2\omega_0\,t - 1\right)dt$$

另一方面：$L = p\,\dot{q} - H = E\left(\sin^2\omega_0\,t - \cos^2\omega_0\,t\right) \quad \Longrightarrow \quad U(p,\,Q,\,t) = -\int L\,dt$

注意当写成 $U(q,\,P,\,t)$ 时，$\quad U(q,\,P,\,t) = \int L\,dt$

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Apply the method of the action and angle variables to determine the frequencies of a three-dimensional harmonic oscillator with pairwise different force constants. Does the system perform the periodic motion?

Solution

$$H = \frac{1}{2\,m}\left(p_x^2 + p_y^2 + p_z^2\right) + \frac{m}{2}\left(\omega_1^2\,x^2 + \omega_2^2\,y^2 + \omega_3^2\,z^2\right)$$

$$S(q,\,P,\,t) = -E\,t + W(q,\,P)$$

哈雅方程：$\dfrac{1}{2\,m}\left[\left(\dfrac{\partial W}{\partial x}\right)^2 + \left(\dfrac{\partial W}{\partial y}\right)^2 + \left(\dfrac{\partial W}{\partial z}\right)^2\right] + \dfrac{m}{2}\left(\omega_1^2\,x^2 + \omega_2^2\,y^2 + \omega_3^2\,z^2\right) = E$

分离变量：$W = W_1(x,\,P) + W_1(y,\,P) + W_1(z,\,P)$

$$\begin{cases} \dfrac{1}{2\,m}\left(\dfrac{dW_3}{dz}\right)^2 + \dfrac{m}{2}\,\omega_3^2\,z^2 = \alpha_z > 0 \\ \dfrac{1}{2\,m}\left(\dfrac{dW_2}{dy}\right)^2 + \dfrac{m}{2}\,\omega_2^2\,y^2 = \alpha_y > 0 \\ \dfrac{1}{2\,m}\left(\dfrac{dW}{dx}\right)^2 + \dfrac{m}{2}\,\omega_1^2\,x^2 = E - \alpha_y - \alpha_z > 0 \end{cases} \qquad \text{实际上就是} \begin{cases} \dfrac{p_z^2}{2\,m} + \dfrac{m}{2}\,\omega_3^2\,z^2 = \alpha_z \\ \dfrac{p_y^2}{2\,m} + \dfrac{m}{2}\,\omega_2^2\,y^2 = \alpha_y \\ \dfrac{p_x^2}{2\,m} + \dfrac{m}{2}\,\omega_1^2\,x^2 = \alpha_x = E - \alpha_y - \alpha_z \end{cases}$$

桑额相平面均为天平动，轨迹均为椭圆，三个面积分别是

$$J_z = \frac{1}{2\,\pi}\oint p_z\,dz = \frac{1}{2}\sqrt{2\,m\,\alpha_z}\;\sqrt{2\,\alpha_z / m\,\omega_3^2} = \frac{\alpha_z}{\omega_3}$$

$$J_y = \frac{1}{2\,\pi}\oint p_y\,dy = \frac{1}{2}\sqrt{2\,m\,\alpha_y}\;\sqrt{2\,\alpha_y / m\,\omega_2^2} = \frac{\alpha_y}{\omega_2}$$

$$J_x = \frac{1}{2\,\pi}\oint p_x\,dx = \frac{1}{2}\sqrt{2\,m\,\alpha_x}\;\sqrt{2\,\alpha_x / m\,\omega_3^2} = \frac{\alpha_x}{\omega_1}$$

$$E = \alpha_x + \alpha_y + \alpha_z \quad \Longrightarrow \quad E = \omega_1\,J_x + \omega_2\,J_y + \omega_3\,J_z$$

$\omega_x = \dfrac{\partial E}{\partial J_x} = \omega_1$，类似地：$\omega_1 = \dfrac{\partial E}{\partial J_y} = \omega_2$，$\quad \omega_3 = \dfrac{\partial E}{\partial J_z} = \omega_3$

三个角频率分别为：ω_1，ω_2，和 ω_3

当 $\dfrac{\omega_1}{\omega_2}$ 和 $\dfrac{\omega_3}{\omega_2}$ 均为有理数时，体系时是周期的，否则是多周期体系。

★ 参照 课件第六章 例题 6.2.09 及文献 **American Journal of Physics 82, 848 (2014)** （已上传）求解 （荣誉课程同学完成）

一个电量为 q 的点电荷在固定于坐标原点，电偶极矩为 $\vec{p} = p_0 \vec{e}_z$ 的点电偶极子产生的静电场中，

点电荷的静电势能为 $V(r, \theta) = \dfrac{k \cos\theta}{r^2}$，其中 r, θ, ϕ 为球坐标，$k = \dfrac{q\, p_0}{4\pi\varepsilon_0}$。

(a) 写出体系的拉格朗日量及对应于三个广义坐标 r, θ, ϕ 的拉格朗日方程

(b) 写出对应于循环坐标 ϕ 的运动常数 ℓ，并解释其物理意义

(c) 求点电荷在 $\ell = 0,\ \phi = 0,\ r = r_0$ 条件下的解，并求此解对应的能量 E，其中 r_0 是常数

(d) 写出体系的哈雅方程，并求哈密顿主函数 $S(r, \theta, \phi, E, \alpha_\theta, \alpha_\phi)$

(e) 讨论 E > 0, E = 0, E < 0 三种情况下的解 r(t)，只需给出 r(t) 的函数形式，不必求出常数。

(f) 试判断求解哈雅方程能否得到求解拉格朗日方程 (c) 所求出的解。

若不能，试解释哈雅方程为什么会漏掉这个解。

Solution

一步步的解答见 Am. J. Phys. **82**, 848 (2014)